

The Writer Wins, the Emitter Learns: Same-Day Answers to the Open Questions of the Composition and Cadence Notes

Franci Penov
kortexa.ai
francip@kortexa.ai

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Abstract

The composition note [1] and the cadence note [2] each ended by naming its own missing experiment: M3, the registered depth-partition control that would separate “one writer” from “writers that merely stopped colliding in parameters,” and C2b, the registered re-encoding consistency emitter aimed at the cadence loop’s one located failure. This note reports both, run the same day, each landing on the side its parent note predicted. M3: the creature pair’s two blocks, retrained as independent single-lane bridges on disjoint halves of the 4B host’s eight full-attention layers (393,216 parameters each, zero overlap, disjointness verified by identity against the armed tower), are perfect solo (1.00 at every seed) — and every merged cell sits at the floor: joint 0.000 at all three seeds, against the one-writer bridge’s 1.00/0.90/1.00 on the same items, with the wreckage a different dominance pattern each seed. Composition dies in the shared residual stream, not in parameter collisions; the one-writer attribution stands. C2b: one consistency term — the packet encoded from a goal’s canonical acted pass must reproduce the packet encoded from its assignment exchange — and the C1 gate still passes at every seed (0.90/0.80/1.00 against 0.20 floors) while the re-run C2 loop returns reading (i) unanimously: live re-emitted goal-consistency over hops 4–7 rises from 0.20 flat to 0.80/0.60/1.00, matching each seed’s own frozen arm, one seed perfect across all sixteen ticks with zero contamination. The mechanism the program was building toward is now measured end to end at creature scale: one writer speaks for many sensors, and a goal the model writes itself survives its own cadence.

1 Two notes, two named gaps

Paper 11 established that a single hypernetwork bridge reading N sensors composes where every merged-writer route had died, but declined to attribute the win between “one writer” and “natively-trained joint data” — M3, registered beside M1, was named as the experiment that separates them. Paper 12 established that a goal packet endures sixteen ticks when replayed frozen but dies on its first re-encoding, located the failure in the emitter’s training distribution, and named C2b — registered before anything ran — as the follow-up. Both experiments ran on 2026-08-09, the day their parent notes were published. This note is the pair of answers.

2 M3: writers that cannot collide still fail

The design removes parameter interaction entirely. The `temporal+vitals` blocks were retrained from scratch as independent single-lane bridges — the M1 recipe, one lane each, no joint data

seed	solo A	solo B	merge joint	merge A/B	shuffled	M1 joint	fit A/B
42	1.00	1.00	0.000	0.20/0.30	0.000	1.00	0.86/0.95
1042	1.00	1.00	0.000	0.40/0.10	0.000	0.90	0.89/0.96
2042	1.00	1.00	0.000	0.10/0.70	0.000	1.00	0.90/0.96

Table 1: M3 at 4B, **temporal+vitals**: depth-partitioned independent writers, merged by concatenation. Every seed is valid (solo retention perfect) and every merge sits at the floor the one-writer bridge cleared on the same items.

seed	C1 gate	live 4–7		live 0–3	post-switch	frozen 4–15	contam.
		C2	C2b				
42	0.90	0.20	0.85	0.80	0.80	1.00	8/40
1042	0.80	0.20	0.60	0.60	0.65	0.60	7/40
2042	1.00	0.20	1.00	1.00	1.00	1.00	0/40

Table 2: C2b at 230M: the C1 gate re-check and the re-run C2 loop. The live 4–7 column pairs each seed’s C2 baseline with its C2b value; the frozen column is the carrier ceiling the live arm now matches.

— with LoRA targets on disjoint halves of the eight full-attention layers: lane A on layers 3, 7, 11, 15, lane B on layers 19, 23, 27, 31, 393,216 parameters per half, tiling the original channel exactly. Merging is concatenation: each block owns its layers outright, and the harness verifies the disjointness by identity against the armed tower rather than assuming it from a path convention.

The result (Table 1, Figure 1 left) is unambiguous. Solo, both halves are perfect: 1.00 A-heed and 1.00 B-heed at every seed — a half-depth channel loses nothing on its own lane. Merged, joint-heed is **0.000 at every seed**, and the per-lane rates collapse to a dominance pattern with a different winner each seed (0.20/0.30; 0.40/0.10; 0.10/0.70) — both lanes degrade, neither survives whole, the joint never fires. The state-deranged shuffled merge sits at 0.000, and the router-oracle on the retrained solos is 0.000 — selection still cannot make one utterance satisfy two scorers. This is paper 7’s layer-partition lesson reproduced at the M1 venue with the M1 recipe: zero weight-space interference buys nothing, because composition dies in the shared residual stream — there is one stage no matter how the writers are wired. **The attribution closes: M1’s win is the natively-trained single writer.**

3 C2b: one consistency term closes the emitter

C2’s diagnosis was exact: the emitter (the layer-12 tap plus the bridge’s encoder) was trained only on assignment-exchange surfaces, and the loop hands it *acted* passes — the serve-time action menu plus a JSON tool call — which it encodes off-distribution, losing the goal in one hop. C2b, registered before anything ran, adds exactly what the diagnosis asks: for each training item with goal G , the canonical acted surface under G is also encoded, and the training gains $1 - \cos(\text{enc}(\text{acted-}G), \text{sg}[\text{enc}(\text{assignment-}G)])$ plus the auxiliary goal classifier applied to the acted packet — the assignment side stop-graduated so the original regime stays the anchor. Everything else is C1-verbatim, and the protocol re-checks the C1 gate before any loop runs: the consistency term must not buy loop stability with C1’s currency.

It does not (Table 2): goal-consistent choice on held-out phrasings is 0.90/0.80/1.00 against the same three floors C1 measured (no packet 0.20, undirected waking 0.20, wrong-goal packet above

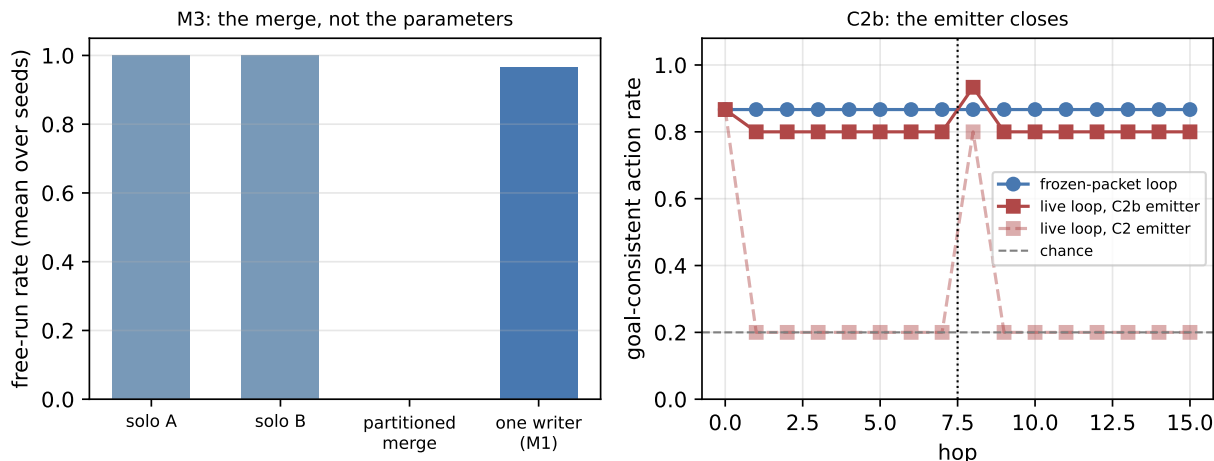


Figure 1: The two answers. Left: M3 at 4B — depth-partitioned writers are perfect solo and dead merged, while the one-writer bridge composes on the same items. Right: C2b at 230M, pooled over five goals and three seeds — the same sixteen-tick loop that collapsed at hop 1 under the C2 emitter (faded) holds at the frozen arm’s level under the consistency-trained emitter; the dotted line marks the scripted goal switch.

both), zero-dose bit-exact, base weights byte-identical. And the loop, re-run verbatim, returns **reading (i) at every seed** (Figure 1 right): the live re-emitted arm holds at 0.80/0.60/1.00 over hops 4–7 where the C2 emitter sat at 0.20 flat, the hop-1 collapse is gone, and at every seed the live arm now matches its own frozen arm — seed 2042 runs all sixteen ticks at 1.00 with zero contamination, and seed 1042’s weaker 0.60 is exactly its frozen ceiling, so the re-encoding itself now costs nothing. Post-switch consistency is 0.80/0.65/1.00 with contamination 0–8 of 40. The registered caveat travels with the result: the canonical acted surface covers goal-consistent replies, so off-policy passes — a tick that acts wrong or emits malformed JSON — remain the named residual, and the on-policy question belongs to the tool-pursuit line.

4 What stands now

The two parent notes’ claims compose. Multi-sensor conditioning of a frozen host is solved by one trained mouth-piece, and M3 shows the mouth-piece — not the absence of parameter collisions — is what does the work. A goal in 256 floats directs action, survives frozen replay indefinitely, and now survives its own re-encoding: the creature-scale loop — write the want, act under it, re-write it from the acted pass, tick after tick, with a clean scripted switch — is measured working end to end. What was a research gap this morning is a deployment decision tonight, and that decision is held for a human by standing rule. The open edges are the registered ones: on-policy acted surfaces, N beyond three, and the inserted-capacity control (M2).

5 Reproducibility

Artifacts `multi-sensor-m3-4b` and `own-cadence-c2{,b}-230m` are bundled beside this note; every number and the figure render from them alone. Harnesses: `multi_sensor.py --stage m3` and

`own_cadence.py --stage c2b`, in `tracks/multi-sensor/` and `tracks/own-cadence/`; registrations and results, each dated before its run, in the tracks' `PLAN.md` files. Hosts: frozen Qwen3.5-4B (M3, CUDA) and frozen LFM2.5-230M (C2b, MPS float32).

References

- [1] F. Penov. The Third Block Composes: One-Writer Bridges for Multi-Sensor Conditioning of a Frozen Language Model. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [2] F. Penov. The Carrier Holds, the Emitter Leaks: A Goal Packet on Its Own Cadence at Creature Scale. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [3] F. Penov. Zero Weight-Space Interference Is Not Enough. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [4] F. Penov. One Mouth, One Mode. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.